

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

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**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
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CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by ARPITA GANGADHAR PATTAR (**1BM24CS053**), who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - (**23CS4PCADA**) work prescribed for the said degree.

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Index Sheet

Sl. No.	Experiment Title	Page No.
1	Write program to obtain the Topological ordering of vertices in a given digraph.	4-6
2	Implement Johnson Trotter algorithm to generate permutations.	7-10
3	Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.	11-13
4	Sort a given set of N integer elements using Quick Sort technique and compute its time taken.	14-16
5	Sort a given set of N integer elements using Heap Sort technique and compute its time taken.	17-19
6	Implement 0/1 Knapsack problem using dynamic programming. (5 Marks)	20-21
7	Implement All Pair Shortest paths problem using Floyd's algorithm.	22-23
8	(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm. (b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.	24-26 27-29
9	Implement Fractional Knapsack using Greedy technique.	30-32
10	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	33-35
11	Implement "N-Queens Problem" using Backtracking.	36-38
12	Leetcode	39-44

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1: Write program to obtain the Topological ordering of vertices in a given digraph.

```
#include <stdio.h>
```

```
void topologicalSort(int n, int adj[100][100])
```

```
{  
    int queue[100];  
    int indegree[100];  
    int front = 0, rear = -1;  
    for (int i = 0; i < n; i++)  
    {  
        indegree[i] = 0;  
    }  
    for (int i = 0; i < n; i++)  
    {  
        for (int j = 0; j < n; j++)  
        {  
            if (adj[i][j] == 1)  
            {  
                indegree[j]++;  
            }  
        }  
    }  
    for (int i = 0; i < n; i++)  
    {  
        if (indegree[i] == 0)  
        {  
            queue[++rear] = i;  
        }  
    }  
}
```

```

printf("Topological Order: ");
while (front <= rear)
{
    int node = queue[front++];
    printf("%d ", node);

    for (int i = 0; i < n; i++)
    {
        if (adj[node][i] == 1)
        {
            indegree[i]--;

            if (indegree[i] == 0)
            {
                queue[++rear] = i;
            }
        }
    }
}
printf("\n");
}

int main()
{
    int n;
    int adj[100][100];

    printf("Enter the number of vertices: ");
    scanf("%d", &n);

```

```
printf("Enter the adjacency matrix:\n");

for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        scanf("%d", &adj[i][j]);
    }
}

topologicalSort(n, adj);

return 0;
}
```

OUTPUT:

```
Enter the number of vertices: 5
Enter the adjacency matrix:
0 1 1 0 0
0 0 0 1 1
0 1 0 1 0
0 0 0 0 0
0 0 0 0 0
Topological Order: 0 2 1 3 4

Process returned 0 (0x0)    execution time : 29.539 s
Press any key to continue.
|
```

Lab Program 2: Implement Johnson Trotter algorithm to generate permutations.

```
#include <stdio.h>
```

```
int LR = 1, RL = 0;
```

```
void swap(int *a, int *b)
```

```
{
```

```
    int temp = *a;
```

```
    *a = *b;
```

```
    *b = temp;
```

```
}
```

```
int fact(int n)
```

```
{
```

```
    int result = 1;
```

```
    for (int i = 1; i <= n; i++)
```

```
        result = result * i;
```

```
    return result;
```

```
}
```

```
int permutation(int a[], int dir[], int n)
```

```
{
```

```
    int prev = 0, mobile = 0, pos = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        if (i != 0 && dir[a[i] - 1] == RL)
```

```
        {
```

```
            if (a[i] > a[i - 1] && a[i] > prev)
```

```
            {
```

```

        mobile = a[i];
        prev = mobile;
    }
}
if (i != n - 1 && dir[a[i] - 1] == LR)
{
    if (a[i] > a[i + 1] && a[i] > prev)
    {
        mobile = a[i];
        prev = mobile;
    }
}
}
for (int i = 0; i < n; i++)
{
    if (a[i] == mobile)
        pos = i + 1;
}
if (dir[a[pos - 1] - 1] == RL)
    swap(&a[pos - 1], &a[pos - 2]);
else if (dir[a[pos - 1] - 1] == LR)
    swap(&a[pos], &a[pos - 1]);

for (int i = 0; i < n; i++)
{
    if (a[i] > mobile)
    {

```



```

        if (dir[a[i] - 1] == LR)
            dir[a[i] - 1] = RL;
        else if (dir[a[i] - 1] == RL)
            dir[a[i] - 1] = LR;
    }
}

for (int i = 0; i < n; i++)
    printf("%d ", a[i]);
printf("\n");
return 0;
}

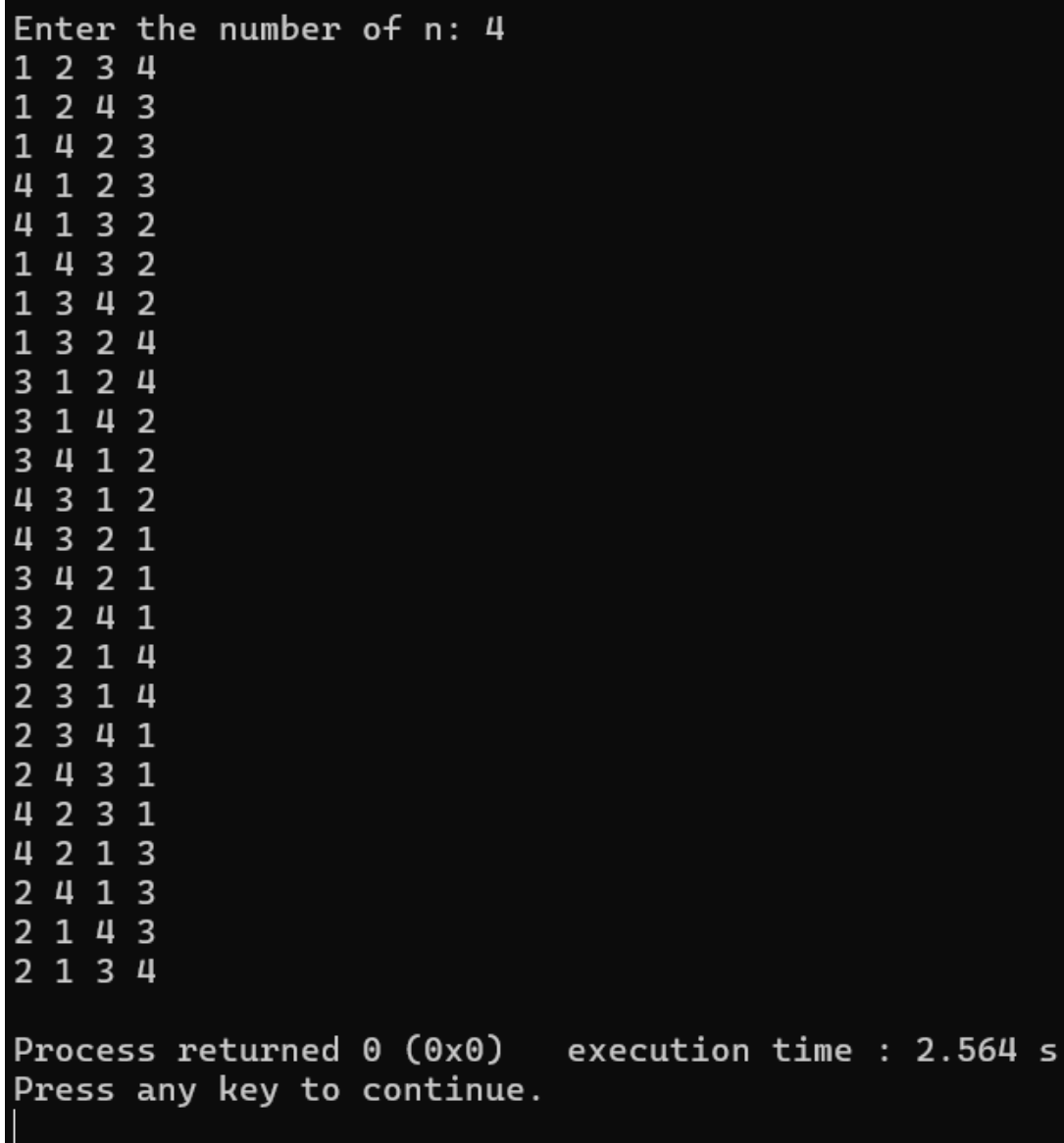
int main()
{
    int n;
    printf("Enter the number of n: ");
    scanf("%d", &n);
    int a[n], dir[n];

    for (int i = 0; i < n; i++)
    {
        a[i] = i + 1;
        printf("%d ", a[i]);
    }
    printf("\n");
}

```

```
for (int i = 0; i < n; i++)  
    dir[i] = RL;  
for (int i = 1; i < fact(n); i++)  
    permutation(a, dir, n);  
  
return 0;  
}
```

Ouput:



```
Enter the number of n: 4  
1 2 3 4  
1 2 4 3  
1 4 2 3  
4 1 2 3  
4 1 3 2  
1 4 3 2  
1 3 4 2  
1 3 2 4  
3 1 2 4  
3 1 4 2  
3 4 1 2  
4 3 1 2  
4 3 2 1  
3 4 2 1  
3 2 4 1  
3 2 1 4  
2 3 1 4  
2 3 4 1  
2 4 3 1  
4 2 3 1  
4 2 1 3  
2 4 1 3  
2 1 4 3  
2 1 3 4  
  
Process returned 0 (0x0)   execution time : 2.564 s  
Press any key to continue.  
|
```

Lab Program 3: Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.

```
#include<stdio.h>
```

```
void MergeSort(int a[],int b[],int m,int n)
```

```
{
```

```
    int i=0,j=0,k=0,c[100];
```

```
    while(i<m && j<n)
```

```
    {
```

```
        if(a[i]<b[j])
```

```
        {
```

```
            c[k]=a[i];
```

```
            k++;
```

```
            i++;
```

```
        }
```

```
    else
```

```
    {
```

```
        c[k]=b[j];
```

```
        k++;
```

```
        j++ ;
```

```
    }
```

```
}
```

```
while(i<m)
```

```
{
```

```
    c[k]=a[i];
```

```
    k++;
```

```
    i++;
```

```

    }
    while(j<n)
    {
        c[k]=b[j];
        k++;
        j++;
    }
    printf("Merged Array:");
    for(int i=0;i<(m+n);i++)
        printf("%d\t",c[i]);
}

int main()
{
    int n,m,a[100],b[100];
    printf("Enter the number of elements in First array:");
    scanf("%d",&m);
    printf("Enter the number of elements of first array:");
    for(int i=0;i<m;i++)
        scanf("%d",&a[i]);

    printf("Enter the number of elements in Second array:");
    scanf("%d",&n);
    printf("Enter the number of elements of first array:");
    for(int i=0;i<n;i++)
        scanf("%d",&b[i]);

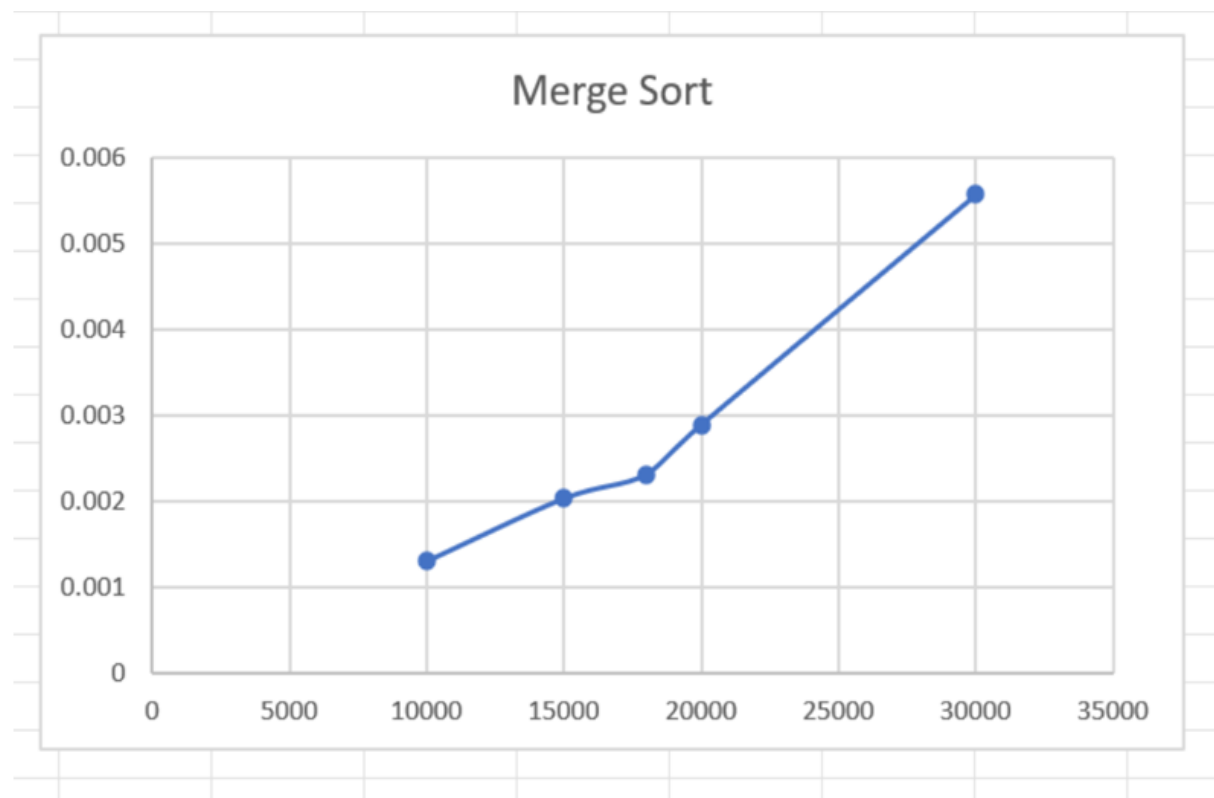
```

```
MergeSort(a,b,m,n);  
  
return 0;  
  
}
```

Output:

```
Enter the number of elements in First array:5  
Enter the number of elements of first array:1 3 5 7 9  
Enter the number of elements in Second array:5  
Enter the number of elements of first array:2 4 6 8 10  
Merged Array:1 2 3 4 5 6 7 8 9 10  
Process returned 0 (0x0) execution time : 15.473 s  
Press any key to continue.  
|
```

Graph:



Lab Program 4: Sort a given set of N integer elements using Quick Sort technique and compute its time taken.

```
#include<stdio.h>
```

```
int partition(int a[] ,int low,int high)
```

```
{
```

```
    int i,j,temp,key;
```

```
    i=low+1;
```

```
    j=high;
```

```
    key=a[low];
```

```
    while(1)
```

```
    {
```

```
        while(a[i]<=key && i<=high)
```

```
            i++;
```

```
        while(a[j]>key)
```

```
            j--;
```

```
        if(i<j){
```

```
            temp=a[i];
```

```
            a[i]=a[j];
```

```
            a[j]=temp;
```

```
        }
```

```
    else{
```

```
        temp=a[low];
```

```
        a[low]=a[j];
```

```
        a[j]=temp;
```

```
        return j;
```

```
    }
```

```
}
```

```
}
```

```

void QuickSort(int a[],int low,int high)
{
    int j;
    if(low<high)
    {
        j=partition(a,low,high);
        QuickSort(a,low,j-1);
        QuickSort(a,j+1,high);
    }
}

int main()
{
    int n,a[100];
    printf("Enter the number of elements in the array:");
    scanf("%d",&n);
    printf("Enter the array elements:");
    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);
    QuickSort(a,0,n-1);

    printf("Sorted Array:");
    for(int i=0;i<n;i++)
        printf("%d\t",a[i]);

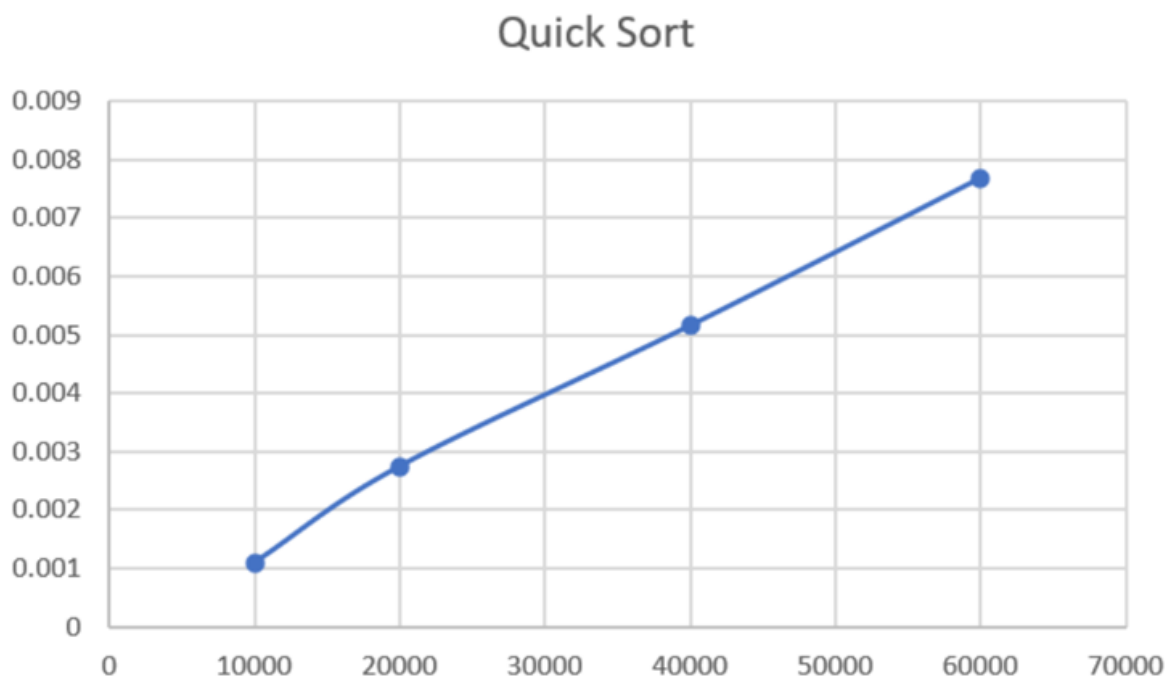
    return 0;
}

```

Output:

```
Enter the number of elements in the array:6
Enter the array elements:12 5 8 23 56 89
Sorted Array:5 8 12 23 56 89
Process returned 0 (0x0) execution time : 11.668 s
Press any key to continue.
|
```

Graph:



Lab Program 5: Sort a given set of N integer elements using Heap Sort technique and compute its time taken.

```
#include<stdio.h>
```

```
void heapify(int a[],int n,int p)
```

```
{
```

```
    int c,item;
```

```
    item=a[p];
```

```
    c=2*p+1;
```

```
    while(c<n)
```

```
    {
```

```
        if(c+1<n)
```

```
        {
```

```
            if(a[c]<a[c+1])
```

```
                c++;
```

```
        }
```

```
        if(item<a[c])
```

```
        {
```

```
            a[p]=a[c];
```

```
            p=c;
```

```
            c=2*p+1;
```

```
        }
```

```
        else
```

```
            break;
```

```
    }
```

```
    a[p]=item;
```

```
}
```

```

void Build_heap(int a[],int n)
{
    for(int i=(n-1)/2;i>=0;i--)
        heapify(a,n,i);
}

void heapsort(int a[],int n)
{
    int i,temp;
    Build_heap(a,n);

    for(int i=n-1;i>0;i--)
    {
        temp=a[0];
        a[0]=a[i];
        a[i]=temp;
        heapify(a,i,0);
    }
    printf("Sorted Array:");
    for(int i=0;i<n;i++)
        printf("%d\t",a[i]);
}

int main()
{
    int a[100],n,i;
    printf("Enter the number of elements:");
    scanf("%d",&n);
    printf("Enter the array elements: ");
    for(i=0;i<n;i++)

```

```
        scanf("%d",&a[i]);  
    heapsort(a,n);  
    return 0;  
}
```

Output:

```
Enter the number of elements:8  
Enter the array elements: 13 8 5 2 1 19 26 15  
Sorted Array:1  2      5      8      13      15      19      26  
Process returned 0 (0x0)   execution time : 24.709 s  
Press any key to continue.  
|
```

Lab Program 6: Implement 0/1 Knapsack problem using dynamic programming.

```
#include<stdio.h>
```

```
int max(int a,int b)
```

```
{  
    if(a>b)  
        return a;  
    return b;  
}
```

```
int main()
```

```
{  
    int p[100],w[100],M,n,table[100][100];  
    printf("Enter the number of items:");  
    scanf("%d",&n);  
    printf("Enter the maximum capacity of Knapsack:");  
    scanf("%d",&M);  
    for(int i=1;i<=n;i++){  
        printf("Enter the weight and profit of item %d:",i);  
        scanf("%d %d",&w[i],&p[i]);  
    }  
    for(int i=0;i<=n;i++){  
        for(int j=0;j<=M;j++){  
            if(i==0 || j==0) {  
                table[i][j]=0;  
            }  
            else if(j<w[i]) {  
                table[i][j]=table[i-1][j];  
            }  
        }  
    }  
}
```

```
        else {  
            table[i][j]=max(table[i-1][j],p[i]+table[i-1][j-w[i]]);  
        }  
    }  
    printf("Maximum profit made:%d",table[n][M]);  
    return 0;  
}
```

Output:

```
Enter the number of items:3  
Enter the maximum capacity of Knapsack:50  
Enter the weight and profit of item 1:10 60  
Enter the weight and profit of item 2:20 100  
Enter the weight and profit of item 3:30 120  
Maximum profit made:220  
Process returned 0 (0x0)   execution time : 16.977 s  
Press any key to continue.  
|
```

Lab Program 7: Implement All Pair Shortest paths problem using Floyd's algorithm.

```
#include<stdio.h>
```

```
#define INF 99
```

```
void Floyd's(int a[10][10],int n)
```

```
{ for(int k=0;k<n;k++)
```

```
{
```

```
for(int i=0;i<n;i++)
```

```
{
```

```
for(int j=0;j<n;j++)
```

```
if(a[i][k]+a[k][j]<a[i][j])
```

```
a[i][j]=a[i][k]+a[k][j];
```

```
}
```

```
}
```

```
printf("Shortest Distance Matrix:\n");
```

```
for(int i=0;i<n;i++)
```

```
{
```

```
for(int j=0;j<n;j++)
```

```
if(a[i][j]==INF)
```

```
printf("INF");
```

```
else
```

```
printf("%d ",a[i][j]);
```

```
printf("\n");
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
int i,j,k,n;
```

```

int cost[10][10],a[10][10];

printf("Enter the number of Vertices:");

scanf("%d",&n);

printf("Enter the cost matrix:\n");

for(i=0;i<n;i++)
{
    for(j=0;j<n;j++)
    {
        scanf("%d",&cost[i][j]);

        a[i][j]=cost[i][j];
    }
}

Floyds(a,n);

return 0;
}

```

Output:

```

Enter the number of Vertices:5
Enter the cost matrix:
0 4 99 5 99
99 0 1 99 6
2 99 0 3 99
99 99 1 0 2
1 99 99 4 0
Shortest Distance Matrix:
0 4 5 5 7
3 0 1 4 6
2 6 0 3 5
3 7 1 0 2
1 5 5 4 0

Process returned 0 (0x0)    execution time : 36.980 s
Press any key to continue.
|

```

Lab Program 8:(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.

```
#include <stdio.h>

#include <time.h>

#define MAX INT_MAX

int minKey(int key[], int set[], int n)
{
    int min = MAX, min_id;
    for (int i = 0; i < n; i++)
    {
        if (set[i] == 0 && key[i] < min)
        {
            min = key[i];
            min_id = i;
        }
    }
    return min_id;
}

int main()
{
    int n, parent[100], key[100], set[100], graph[100][100];
    int mincost = 0;
    clock_t start, end;
    printf("Enter the number of vertices in the graph: ");
    scanf("%d", &n);
    printf("Enter the adjacency matrix:\n");
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
```



```

        scanf("%d", &graph[i][j]);
    }
}
start = clock();

for (int i = 0; i < n; i++)
{
    key[i] = MAX;
    set[i] = 0;
}
key[0] = 0;
parent[0] = -1;

for (int count = 0; count < n - 1; count++)
{
    int i = minKey(key, set, n);
    set[i] = 1;
    for (int j = 0; j < n; j++)
    {
        if (graph[i][j] &&
            set[j] == 0 &&
            graph[i][j] < key[j])
        {
            parent[j] = i;
            key[j] = graph[i][j];
        }
    }
}
end = clock();
printf("\nEdge\tWeight\n");
for (int i = 1; i < n; i++)
{
    printf("%d - %d\t%d\n",parent[i],i, graph[parent[i]][i]);
}

```

```

        mincost += graph[parent[i]][i];
    }
    printf("\nMinimum cost is %d\n", mincost);
    double time = (double)(end - start) / CLOCKS_PER_SEC;
    printf("Execution time: %f\n", time);
    return 0;
}

```

Output:

```

Enter the number of vertices in the graph: 5
Enter the adjacency matrix:
0 2 0 6 0
2 0 3 8 5
0 3 0 0 7
6 8 0 0 9
0 5 7 9 0

Edge      Weight
0 - 1     2
1 - 2     3
0 - 3     6
1 - 4     5

Minimum cost is 16
Execution time: 0.000000

Process returned 0 (0x0)    execution time : 50.840 s
Press any key to continue.

```

Lab Program 8:(b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.

```
#include<stdio.h>
```

```
typedef struct
```

```
{  
    int u,v,w;  
}Edge;
```

```
Edge edges[100],result[100];
```

```
int parent[100];
```

```
int find(int i)
```

```
{  
    while(parent[i]!=i)  
    {  
        i=parent[i];  
    }  
    return i;  
}
```

```
void unionSet(int i,int j)
```

```
{  
    parent[i]=j;  
}
```

```
void sortEdge(int e)
```

```
{  
    int i,j;  
    Edge temp;  
    for(i=0;i<e-1;i++){  
        for(j=0;j<e-1-i;j++){  
            {
```

```

        if(edges[j].w>edges[j+1].w){
            temp=edges[j];
            edges[j]=edges[j+1];
            edges[j+1]=temp;
        }
    }
}

int kruskals(int n,int e)
{
    int count=0;
    int cost=0;
    int u,v;
    for(int i=0;i<n;i++){
        parent[i]=i;
    }
    sortEdge(e);
    for(int i=0;i<e&& count<n-1;i++){
        {
            u=find(edges[i].u);
            v=find(edges[i].v);
            if(u!=v){
                result[count++]=edges[i];
                cost+=edges[i].w;
                unionSet(u,v);
            }
        }
    }
    printf("Minimum Spanning Tree:\n");
    for(int i=0;i<count;i++){
        printf("%d---%d = %d\n",result[i].u,result[i].v,result[i].w);
    }
}

```

```
    return cost;
}
```

Output:

```
Enter the number of vertices:7
Enter the number of edges:9
Enter the edges(u,v,weight):
1 2 4
1 3 28
2 5 26
3 6 12
3 4 8
4 5 22
4 7 16
6 7 6
5 7 20
Minimum Spanning Tree:
1---2 = 4
6---7 = 6
3---4 = 8
3---6 = 12
5---7 = 20
2---5 = 26
Total Cost:76

Process returned 0 (0x0)    execution time : 55.094 s
Press any key to continue.
|
```

Lab Program 9: Implement Fractional Knapsack using Greedy technique.

```
#include <stdio.h>
```

```
int main() {  
    int i, j, n, m;  
    int p[50], w[50];  
    float x[50];  
    int t1, t2;  
  
    printf("Enter the number of objects: ");  
    scanf("%d", &n);  
    printf("Enter the weights of all objects:\n");  
    for (i = 0; i < n; i++) {  
        scanf("%d", &w[i]);  
    }  
    printf("Enter the profit of all objects:\n");  
    for (i = 0; i < n; i++) {  
        scanf("%d", &p[i]);  
    }  
    for (i = 0; i < n - 1; i++) {  
        for (j = i + 1; j < n; j++) {  
            if ((float)p[i] / w[i] < (float)p[j] / w[j]) {  
  
                t1 = p[i];  
                p[i] = p[j];  
                p[j] = t1;  
  
                t2 = w[i];
```

```

        w[i] = w[j];
        w[j] = t2;
    }
}

printf("Enter the maximum capacity: ");
scanf("%d", &m);

for (i = 0; i < n; i++) {
    x[i] = 0.0;
}

int rem_cap = m;
for (i = 0; i < n; i++) {
    if (w[i] > rem_cap) {
        break;
    }
    x[i] = 1.0;
    rem_cap -= w[i];
}

if (i < n) {
    x[i] = (float)rem_cap / w[i];
}

float sum = 0;
for (i = 0; i < n; i++) {
    sum = sum + (p[i] * x[i]);
}

printf("Total profit : %f\n", sum);

return 0;
}

```

Output:

```
Enter the number of objects: 3
Enter the weights of all objects:
18 15 10
Enter the profit of all objects:
25 24 15
Enter the maximum capacity: 20
Total profit : 31.500000

Process returned 0 (0x0)   execution time : 13.103 s
Press any key to continue.
```


Lab Program 10: From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm

```
#include <stdio.h>
```

```
#include <time.h>
```

```
#define n 5
```

```
int minDist(int dist[], int visited[]) {  
    int min = 99999, min_index = -1;  
    for (int i = 0; i < n; i++) {  
        if (visited[i] == 0 && dist[i] <= min) {  
            min = dist[i];  
            min_index = i;  
        }  
    }  
    return min_index;  
}
```

```
int dijkstra(int weight[n][n], int src) {  
    int dist[n];  
    int visited[n];  
    for (int i = 0; i < n; i++) {  
        dist[i] = 99999;  
        visited[i] = 0;  
    }  
    dist[src] = 0;  
    for (int count = 0; count < n - 1; count++) {  
        int u = minDist(dist, visited);  
        visited[u] = 1;  
        for (int v = 0; v < n; v++) {
```

```

        if (!visited[v] && weight[u][v] != 0 && dist[u] != 99999) {
            int new_dist = dist[u] + weight[u][v];
            if (new_dist < dist[v]) {
                dist[v] = new_dist;
            }
        }
    }
}

printf("Vertex \t Distance from source\n");
for (int i = 0; i < n; i++) {
    printf("%d \t\t %d\n", i + 1, dist[i]);
}

return 0;
}

int main() {
    int i, j;
    int graph[n][n];
    clock_t start, end;

    printf("Enter the weight matrix (%dx%d):\n", n, n);
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &graph[i][j]);
        }
    }

    int source = 0;
    start = clock();
    dijkstra(graph, source);
    end = clock();
}

```

```
double time_taken = (double)(end - start) / CLOCKS_PER_SEC;

printf("Time taken: %f seconds\n", time_taken);

return 0;
}
```

Output:

```
Enter the weight matrix (5x5):
0 10 0 30 100
10 0 50 0 0
0 50 0 20 10
30 0 20 0 60
100 0 10 60 0
Vertex    Distance from source
1          0
2          10
3          50
4          30
5          60
Time taken: 0.001000 seconds

Process returned 0 (0x0)    execution time : 124.393 s
Press any key to continue.
```

Lab Program 11: Implement “N-Queens Problem” using Backtracking.

```
#include <stdio.h>

#include <stdbool.h>

int solutionCount = 0;

bool isSafe(int N, int board[N][N], int row, int col) {
    int i, j;
    for (i = 0; i < col; i++) {
        if (board[row][i]) {
            return false;
        }
    }
    for (i = row, j = col; i >= 0 && j >= 0; i--, j--) {
        if (board[i][j]) {
            return false;
        }
    }
    for (i = row, j = col; j >= 0 && i < N; i++, j--) {
        if (board[i][j]) {
            return false;
        }
    }
    return true;
}

void printSolution(int N, int board[N][N]) {
    printf("Solution %d:\n", ++solutionCount);
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
```

```

        printf("%d ", board[i][j]);
    }
    printf("\n");
}
printf("\n");
}

void solveNQueenRecursive(int N, int board[N][N], int col) {
    if (col >= N) {
        printSolution(N, board);
        return;
    }
    for (int i = 0; i < N; i++) {
        if (isSafe(N, board, i, col)) {
            board[i][col] = 1;
            solveNQueenRecursive(N, board, col + 1);
            board[i][col] = 0;
        }
    }
}

void solveNQ(int N) {
    int board[N][N];
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            board[i][j] = 0;
        }
    }

    solveNQueenRecursive(N, board, 0);

    if (solutionCount == 0) {
        printf("Solution does not exist\n");
    }
}

```

```

    } else {
        printf("Total solutions = %d\n", solutionCount);
    }
}

int main() {
    int N;

    printf("Enter value of N: ");
    scanf("%d", &N);

    if (N <= 0) {
        printf("Enter a positive number.\n");
        return 0;
    }

    solveNQ(N);

    return 0;
}

```

Output:

```

Enter value of N: 4
Solution 1:
0 0 1 0
1 0 0 0
0 0 0 1
0 1 0 0

Solution 2:
0 1 0 0
0 0 0 1
1 0 0 0
0 0 1 0

Total solutions = 2

Process returned 0 (0x0)   execution time : 14.530 s
Press any key to continue.

```

Leetcode: 1)Leet Code exercises on topological sorting(207).

```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int* prerequisitesColSize) {
```

```
    int inDegree[1000] = {0};
```

```
    int queue[1000], front, rear, count;
```

```
    front = rear = count = 0;
```

```
    for (int i = 0; i < prerequisitesSize; i++) {
```

```
        int course = prerequisites[i][0];
```

```
        inDegree[course]++;
```

```
    }
```

```
    for (int i = 0; i < numCourses; i++) {
```

```
        if (inDegree[i] == 0) {
```

```
            queue[rear++] = i;
```

```
        }
```

```
    }
```

```
    while (front < rear) {
```

```
        int currCourse = queue[front++];
```

```
        count++;
```

```
        for (int i = 0; i < prerequisitesSize; i++) {
```

```
            if (prerequisites[i][1] == currCourse) {
```

```
                int depCourse = prerequisites[i][0];
```

```
                inDegree[depCourse]--;
```

```
                if (inDegree[depCourse] == 0) {
```

```
                    queue[rear++] = depCourse;
```

```
                }
```

```
            }
```

```
        }
```

```
    }
```

```
    return count == numCourses;  
}
```

Output:

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

numCourses =
2

prerequisites =
[[1,0]]

Output

true

Expected

true

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

numCourses =
2

prerequisites =
[[1,0],[0,1]]

Output

false

Expected

false

2)Leet Code exercise on Knapsack problem using dynamic Programming(801).

```
int minSwap(int* nums1, int nums1Size, int* nums2, int nums2Size) {  
    int x = 0, y = 1;  
  
    for (int i = 1; i < nums1Size; i++) {  
        int a = x, b = y;  
        if (nums1[i - 1] >= nums1[i] || nums2[i - 1] >= nums2[i]) {  
            x = b;  
            y = a + 1;  
        }  
        else {  
            y = b + 1;  
  
            if (nums1[i - 1] < nums2[i] &&  
                nums2[i - 1] < nums1[i])  
                x = (x < b) ? x : b;  
            y = (y < a + 1) ? y : (a + 1);  
        }  
    }  
    return (x < y) ? x : y;  
}
```

Output:

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

nums1 =
[0,3,5,8,9]

nums2 =
[2,1,4,6,9]

Output

1

Expected

1

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2

Input

nums1 =
[1,3,5,4]

nums2 =
[1,2,3,7]

Output

1

Expected

1

3)Leet Code exercise on Floyd's algorithm(287).

```
int findDuplicate(int* nums, int numsSize)
```

```
{  
    int slow1, fast, slow2;  
    slow1=fast=slow2=0;  
  
    while(true)  
    {  
        slow1=nums[slow1];  
        fast=nums[nums[fast]];  
        if(slow1==fast)  
        {  
            break;  
        }  
    }  
}
```

```
while(true)  
{  
    slow1=nums[slow1];  
    slow2=nums[slow2];  
    if(slow1==slow2)  
    {  
        return slow2;  
    }  
}  
}
```

Output:

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ Case 3

Input

nums =
[1,3,4,2,2]

Output

2

Expected

2

Accepted Runtime: 0 ms

☒ Case 1 ☒ Case 2 ☒ Case 3

Input

nums =
[3,1,3,4,2]

Output

3

Expected

3

☒ Case 1 ☒ Case 2 ☒ Case 3

Input

nums =
[3,3,3,3,3]

Output

3

Expected

3

